

# KIT SAFETY DATA SHEET



**Kit Product Name** ReadyPrep Protein Extraction Kit (Total)

**Kit Catalogue Number(s)** 1632086

**Revision date** 15-Mar-2023

## Kit Contents

| Catalogue Number(s)          | Product Name                              |
|------------------------------|-------------------------------------------|
| 1632101, 1632101EDU, 9703632 | ReadyPrep TBP Reducing Agent              |
| 1632083, 10009795            | ReadyPrep 2-D Rehydration/Sample Buffer 1 |

# SAFETY DATA SHEET

Revision date 23-Aug-2022

Revision Number 1.1

## Section 1: Identification

### Product identifier

**Product Name** ReadyPrep TBP Reducing Agent  
**Catalogue Number(s)** 1632101, 1632101EDU, 9703632

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

### GHS Classification

|                                                  |             |
|--------------------------------------------------|-------------|
| Flammable liquids                                | Category 4  |
| Skin corrosion/irritation                        | Category 2  |
| Serious eye damage/eye irritation                | Category 2  |
| Reproductive toxicity                            | Category 1B |
| Specific target organ toxicity — single exposure | Category 3  |

### Label elements



### **Signal word**

Danger

### **Hazard statements**

Combustible liquid  
Causes skin irritation  
Causes serious eye irritation  
May damage fertility or the unborn child  
May cause respiratory irritation

**Precautionary Statements - Prevention**

Do not handle until all safety precautions have been read and understood

Wear protective gloves/clothing and eye/face protection

Wash face, hands and any exposed skin thoroughly after handling

Avoid breathing dust/fume/gas/mist/vapours/spray

Use only outdoors or in a well-ventilated area

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Skin**

IF ON SKIN: Wash with plenty of water and soap

If skin irritation occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

**Inhalation**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

**Fire**

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

May be harmful if swallowed. May be harmful if inhaled.

**Section 3: Composition/information on ingredients**

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| 1-Methyl-2-pyrrolidone    | 872-50-4    | 50 - 100 |
| Non-hazardous ingredients | Proprietary | Balance  |

**Section 4: First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. IF exposed or concerned: Get medical advice/attention. Get medical attention immediately if symptoms occur.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.

**Skin contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if irritation develops and persists.

**Ingestion**

Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Call a doctor.

**Self-protection of the first aider**

Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Wear personal protective clothing (see section 8). Avoid contact with skin, eyes or clothing.

**Most important symptoms and effects, both acute and delayed**

**Symptoms** May cause redness and tearing of the eyes. Burning sensation.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** Treat symptomatically.

## **Section 5: Fire-fighting measures**

**Suitable Extinguishing Media**

**Suitable Extinguishing Media** Dry chemical. Carbon dioxide (CO<sub>2</sub>). Water spray. Alcohol resistant foam.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

## **Section 6: Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Take precautionary measures against static discharges. Do not touch or walk through spilled material. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** Refer to protective measures listed in Sections 7 and 8. Prevent further leakage or spillage if safe to do so.

**Methods and material for containment and cleaning up**

**Methods for containment** Stop leak if you can do it without risk. Do not touch or walk through spilled material. Dyke far ahead of liquid spill for later disposal.

**Methods for cleaning up** Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## **Section 7: Handling and storage**

**Precautions for safe handling**

**Advice on safe handling** Use personal protection equipment. Do not breathe vapour or mist. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Take precautionary measures against static discharges. Use with local exhaust ventilation. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes. Take off contaminated clothing and wash it before reuse. In case of insufficient ventilation, wear suitable respiratory equipment.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Contaminated work clothing must not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labelled containers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Store locked up. Store according to product and label instructions.

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits**

| Chemical name                      | New Zealand                                                                                      | Australia                                                                                | ACGIH TLV | United Kingdom                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | TWA: 25 ppm<br>TWA: 103 mg/m <sup>3</sup><br>STEL: 75 ppm<br>STEL: 309 mg/m <sup>3</sup><br>Skin | TWA: 25 ppm<br>TWA: 103 mg/m <sup>3</sup><br>STEL: 75 ppm<br>STEL: 309 mg/m <sup>3</sup> | -         | TWA: 10 ppm<br>TWA: 40 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 80 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits**

| Chemical name                      | New Zealand | ACGIH                                                                 |
|------------------------------------|-------------|-----------------------------------------------------------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | -           | 100 mg/L - urine (5-Hydroxy-N-methyl-2-pyrrolidone)<br>- end of shift |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Tight sealing safety goggles.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | Liquid                   |
| Colour          | colourless               |
| Odour           | Amine.                   |
| Odour threshold | No information available |

| <u>Property</u>                        | <u>Values</u>             | <u>Remarks • Method</u> |
|----------------------------------------|---------------------------|-------------------------|
| pH                                     |                           | None known              |
| Melting point / freezing point         | -24 °C                    |                         |
| Boiling point / boiling range          | No data available         | None known              |
| Flash point                            | 90 °C                     |                         |
| Evaporation rate                       | No data available         | None known              |
| Flammability (solid, gas)              | No data available         | None known              |
| Flammability Limit in Air              |                           | None known              |
| Upper flammability or explosive limits | No data available         |                         |
| Lower flammability or explosive limits | No data available         |                         |
| Vapour pressure                        | No data available         | None known              |
| Vapour density                         | No data available         | None known              |
| Relative density                       | No data available         | None known              |
| Water solubility                       | No data available         | Immiscible in water     |
| Solubility(ies)                        | No data available         | None known              |
| Partition coefficient                  | No data available         | None known              |
| Autoignition temperature               | 270 °C                    | None known              |
| Decomposition temperature              |                           | None known              |
| Kinematic viscosity                    | No data available         | None known              |
| Dynamic viscosity                      | No data available         | None known              |
| Explosive properties                   | No information available. |                         |
| Oxidising properties                   | No information available. |                         |

### Other information

|                          |                          |
|--------------------------|--------------------------|
| Softening point          | No information available |
| Molecular weight         | No information available |
| VOC content              | No information available |
| Liquid Density           | No information available |
| Bulk density             | No information available |
| Particle characteristics | No information available |

## Section 10: Stability and reactivity

### Reactivity

|            |                           |
|------------|---------------------------|
| Reactivity | No information available. |
|------------|---------------------------|

### Chemical stability

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

### Explosion data

|                                  |       |
|----------------------------------|-------|
| Sensitivity to mechanical impact | None. |
|----------------------------------|-------|

|                                 |      |
|---------------------------------|------|
| Sensitivity to static discharge | Yes. |
|---------------------------------|------|

### Possibility of hazardous reactions

|                                    |                               |
|------------------------------------|-------------------------------|
| Possibility of hazardous reactions | None under normal processing. |
|------------------------------------|-------------------------------|

### Conditions to avoid

**Conditions to avoid** Heat, flames and sparks.

**Incompatible materials**

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

**Acute toxicity**

**Information on likely routes of exposure**

**Product Information**

|                     |                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. May be harmful if inhaled.                                         |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye irritation. (based on components). May cause redness, itching, and pain.                   |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation. (based on components).                                                                |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea. May be harmful if swallowed. |

**Symptoms** Redness. May cause redness and tearing of the eyes.

**Acute toxicity**

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

|                                      |                 |
|--------------------------------------|-----------------|
| <b>ATEmix (oral)</b>                 | 3,324.90 mg/kg  |
| <b>ATEmix (dermal)</b>               | 26,190.50 mg/kg |
| <b>ATEmix (inhalation-dust/mist)</b> | 5.32 mg/l       |

**Component Information**

| Chemical name          | Oral LD50            | Dermal LD50         | Inhalation LC50        |
|------------------------|----------------------|---------------------|------------------------|
| 1-Methyl-2-pyrrolidone | = 3914 mg/kg ( Rat ) | = 8 g/kg ( Rabbit ) | > 5.1 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes skin irritation.        |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye irritation. |
| <b>Respiratory or skin sensitisation</b> | No information available.                                                              |
| <b>Germ cell mutagenicity</b>            | No information available.                                                              |
| <b>Carcinogenicity</b>                   | No information available.                                                              |

|                                                 |                                                                                                                                                     |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reproductive toxicity</b>                    | Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. May damage fertility or the unborn child. |
| <b>STOT - single exposure</b>                   | May cause respiratory irritation. May cause drowsiness or dizziness.                                                                                |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                                                                           |
| <b>Aspiration hazard</b>                        | No information available.                                                                                                                           |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS.                                                     |

## Section 12: Ecological information

### Ecotoxicity

#### Aquatic ecotoxicity

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name          | Algae/aquatic plants                                  | Fish                                                                                                                                                          | Crustacea                                    |
|------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1-Methyl-2-pyrrolidone | EC50: >500mg/L (72h, <i>Desmodesmus subspicatus</i> ) | LC50: =832mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =1072mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =1400mg/L (96h, <i>Poecilia reticulata</i> ) | EC50: =4897mg/L (48h, <i>Daphnia magna</i> ) |

#### Terrestrial ecotoxicity

| Chemical name          | Earthworm | Avian                                                                 | Honeybees |
|------------------------|-----------|-----------------------------------------------------------------------|-----------|
| 1-Methyl-2-pyrrolidone | -         | Acute Oral Toxicity: LD50 = 2212 mg/kg ( <i>Colinus virginianus</i> ) | -         |

**Persistence and degradability** No information available.

### Bioaccumulative potential

#### Bioaccumulation

#### Component Information

| Chemical name          | Partition coefficient |
|------------------------|-----------------------|
| 1-Methyl-2-pyrrolidone | -0.46                 |

### Mobility in soil

**Mobility** No information available.

### Other adverse effects

No information available.

## Section 13: Disposal considerations

**Waste treatment methods****Waste from residues/unused products**

Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.

Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.

Flammable substances - may not be disposed of into or onto a landfill or sewage facility. They may only be burnt in certain situations.

Flammable gases, liquids and solids may only be discharged into the environment or landfill as waste if the substance will not at any time come into contact with any explosives, oxidising gases, liquids or solids or organic peroxides; and there will be no ignition source in the vicinity of the disposal site at any time and if the substance were to ignite, no person, or place where a person may legally be, would be exposed to an unsafe level of heat radiation.

Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.

Dispose of in accordance with local regulations.

Dispose of waste in accordance with environmental legislation.

**Contaminated packaging**

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.

Packages may only be reused or recycled if:

- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

## Section 14: Transport information

**IATA**

Not regulated

**IMDG**

Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations**

EPA New Zealand HSNO approval code or group standard To be determined

**National regulations**

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below

| Chemical name | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL) |
|---------------|------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|
|---------------|------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|

| Chemical name                      | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL) |
|------------------------------------|------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|
| 1-Methyl-2-pyrrolidone<br>872-50-4 | 0.4 mg/m <sup>3</sup>              | 0.375 mg/L                           | 0.83 mg/m <sup>2</sup>                 | -                                   |

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

**NZIoC**

Contact supplier for inventory compliance status.

**TSCA**

Contact supplier for inventory compliance status.

**DSL/NDSL**

Contact supplier for inventory compliance status.

**EINECS/ELINCS**

Contact supplier for inventory compliance status.

**ENCS**

Contact supplier for inventory compliance status.

**IECSC**

Contact supplier for inventory compliance status.

**KECL**

Contact supplier for inventory compliance status.

**PICCS**

Contact supplier for inventory compliance status.

**AICS**

Contact supplier for inventory compliance status.

#### Legend:

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

## Section 16: Other information

**Revision date**

23-Aug-2022

**Revision Note**

Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

**TWA**

TWA (time-weighted average)

**STEL**

STEL (Short Term Exposure Limit)

**Ceiling**

Maximum limit value

\*

Skin designation

**C**

Carcinogen

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AEGl(s))

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U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 23-Aug-2022

Revision Number 1.1

## Section 1: Identification

### Product identifier

**Product Name** ReadyPrep 2-D Rehydration/Sample Buffer 1

**Catalogue Number(s)** 1632083, 10009795

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

### GHS Classification

|                                 |            |
|---------------------------------|------------|
| <b>Carcinogenicity</b>          | Category 2 |
| <b>Reproductive toxicity</b>    | Category 2 |
| <b>Chronic aquatic toxicity</b> | Category 2 |

### Label elements



### **Signal word**

Warning

### **Hazard statements**

Suspected of causing cancer  
Suspected of damaging fertility or the unborn child  
Toxic to aquatic life with long lasting effects

### **Precautionary Statements - Prevention**

Do not handle until all safety precautions have been read and understood

Wear protective gloves/clothing and eye/face protection  
Avoid release to the environment

**Precautionary Statements - Response**

IF exposed or concerned: Get medical advice/attention

**Spill**

Collect spillage

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

May be harmful if swallowed. May be harmful in contact with skin. Toxic to aquatic life.

**Section 3: Composition/information on ingredients**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

**Section 4: First-aid measures****Description of first aid measures**

|                       |                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | IF exposed or concerned: Get medical advice/attention. Show this safety data sheet to the doctor in attendance.   |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                              |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| <b>Skin contact</b>   | Wash skin with soap and water.                                                                                    |
| <b>Ingestion</b>      | Rinse mouth.                                                                                                      |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**Section 5: Fire-fighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| <b>Large Fire</b> | CAUTION: Use of water spray when fighting fire may be inefficient. |
|-------------------|--------------------------------------------------------------------|

|                                       |                                                                   |
|---------------------------------------|-------------------------------------------------------------------|
| <b>Unsuitable extinguishing media</b> | Do not scatter spilled material with high pressure water streams. |
|---------------------------------------|-------------------------------------------------------------------|

**Specific hazards arising from the chemical**

|                                                   |                           |
|---------------------------------------------------|---------------------------|
| <b>Specific hazards arising from the chemical</b> | No information available. |
|---------------------------------------------------|---------------------------|

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

**Section 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**Section 7: Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store locked up. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

**Section 8: Exposure controls/personal protection****Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Engineering controls | Showers<br>Eyewash stations<br>Ventilation systems. |
|----------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection             | No special protective equipment required.                                                                                                                                |
| Hand protection                 | Wear suitable gloves.                                                                                                                                                    |
| Skin and body protection        | Wear suitable protective clothing.                                                                                                                                       |
| Respiratory protection          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| Environmental exposure controls | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Solid                    |
| Appearance      | solid                    |
| Colour          | white                    |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <b><u>Property</u></b>                 | <b><u>Values</u></b>      | <b><u>Remarks • Method</u></b> |
|----------------------------------------|---------------------------|--------------------------------|
| pH                                     | 10                        |                                |
| Melting point / freezing point         | No data available         | None known                     |
| Boiling point / boiling range          | No data available         | None known                     |
| Flash point                            | No data available         | None known                     |
| Evaporation rate                       | No data available         | None known                     |
| Flammability (solid, gas)              | No data available         | None known                     |
| Flammability Limit in Air              |                           | None known                     |
| Upper flammability or explosive limits | No data available         |                                |
| Lower flammability or explosive limits | No data available         |                                |
| Vapour pressure                        | No data available         | None known                     |
| Vapour density                         | No data available         | None known                     |
| Relative density                       | No data available         | None known                     |
| Water solubility                       | No data available         | Soluble in water               |
| Solubility(ies)                        | No data available         | None known                     |
| Partition coefficient                  | No data available         | None known                     |
| Autoignition temperature               | No data available         | None known                     |
| Decomposition temperature              |                           | None known                     |
| Kinematic viscosity                    | No data available         | None known                     |
| Dynamic viscosity                      | No data available         | None known                     |
| Explosive properties                   | No information available. |                                |
| Oxidising properties                   | No information available. |                                |

**Other information**

|                          |                          |
|--------------------------|--------------------------|
| Softening point          | No information available |
| Molecular weight         | No information available |
| VOC content              | No information available |
| Liquid Density           | No information available |
| Bulk density             | No information available |
| Particle characteristics | No information available |

## Section 10: Stability and reactivity

**Reactivity**

**Reactivity** No information available.

**Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** None known based on information supplied.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

**Acute toxicity**

**Information on likely routes of exposure**

**Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** May be harmful in contact with skin.

**Ingestion** May be harmful if swallowed.

**Symptoms** No information available.

**Acute toxicity**

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 4,300.30 mg/kg

ATEmix (dermal) 2,889.50 mg/kg

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

|                                                 |                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory or skin sensitisation</b>        | No information available.                                                                                                                                      |
| <b>Germ cell mutagenicity</b>                   | No information available.                                                                                                                                      |
| <b>Carcinogenicity</b>                          | Contains a known or suspected carcinogen. Classification based on data available for ingredients. Suspected of causing cancer.                                 |
| <b>Reproductive toxicity</b>                    | Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. Suspected of damaging fertility or the unborn child. |
| <b>STOT - single exposure</b>                   | No information available.                                                                                                                                      |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                                                                                      |
| <b>Aspiration hazard</b>                        | No information available.                                                                                                                                      |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS.                                                                |

## Section 12: Ecological information

### Ecotoxicity

|                                      |                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>           | Toxic to aquatic life. Toxic to aquatic life with long lasting effects.                    |
| <b>Unknown aquatic toxicity</b>      | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>       | There is no data for this product.                                                         |
| <b>Persistence and degradability</b> | No information available.                                                                  |

### Bioaccumulative potential

|                        |                                    |
|------------------------|------------------------------------|
| <b>Bioaccumulation</b> | There is no data for this product. |
|------------------------|------------------------------------|

### Mobility in soil

|                 |                           |
|-----------------|---------------------------|
| <b>Mobility</b> | No information available. |
|-----------------|---------------------------|

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Waste treatment methods

|                                   |                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused</b> | Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 |
|-----------------------------------|-------------------------------------------------------------------------------------------|

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>products</b>               | <p>April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.</p> <p>Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.</p> <p>Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.</p> <p>Environmentally hazardous substances – if the substance, or if it contains a component that is hazardous to the aquatic environment or bioaccumulative and not rapidly degradable, then any component that is bioaccumulative and not rapidly degradable must be removed. The product may only be discharged into the environment if an environmental exposure limit has been set for the substance (or a component of the substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the environmental exposure limit.</p> <p>Dispose of in accordance with local regulations.</p> <p>Dispose of waste in accordance with environmental legislation.</p> |
| <b>Contaminated packaging</b> | <p>For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.</p> <p>Packages may only be reused or recycled if:</p> <ul style="list-style-type: none"> <li>- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;</li> <li>- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Section 14: Transport information

|                               |               |
|-------------------------------|---------------|
| <b><u>IATA</u></b>            | Not regulated |
| <b>UN number or ID number</b> | UN3077        |
| <b>Packing group</b>          | III           |
| <b><u>IMDG</u></b>            | Not regulated |

### Transport in bulk according to Annex II of MARPOL and the IBC Code

No information available

### Special precautions for user

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

**EPA New Zealand HSNO approval code or group standard** To be determined

#### **National regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information  
Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

### **International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

### **International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AICS</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

## **Section 16: Other information**

**Revision date** 23-Aug-2022

**Revision Note** Reformatted and updated existing information.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>*</b>    | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)

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New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**